

Quiz (Carolina Reaper)

- Q1.** Evaluate $f(g(x))$ if $f(x) = x^2$ and $g(x) = 2x + 1$.
(A) $4x^2 + 4x + 1$
(B) $4x^2 + 1$
(C) $2x^2 + 2x + 1$
(D) $4x^2 - 4x + 1$
(E) $2x^2 + 1$
- Q2.** What is the domain of $f(g(x))$ if $f(x) = \frac{1}{x}$ and $g(x) = x - 2$?
(A) All real numbers
(B) All real numbers except $x = 2$
(C) All real numbers except $x = 0$
(D) $x \geq 2$
(E) $x \leq 2$
- Q3.** If $f(x) = x^2$ and $g(x) = |x|$, what is $f(g(-2))$?
(A) 4
(B) -4
(C) 0
(D) 2
(E) -2
- Q4.** A bakery sells a total of x loaves of bread per day. They pack $f(x) = 4x$ loaves in boxes and ship $g(x) = 2x$ boxes. What is $f(g(5))$?
(A) 40
(B) 80
(C) 160
(D) 20
(E) 10
- Q5.** If $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?
(A) $x \geq 2$
(B) $x \leq -2$
(C) $x \geq -2$
(D) $x \leq 2$
(E) $x \geq 0$
- Q6.** What is $g(f(x))$ if $f(x) = 2x$ and $g(x) = x^2 - 1$?
(A) $4x^2 - 1$
(B) $4x^2 + 1$
(C) $2x^2 - 1$
(D) $2x^2 + 1$
(E) $x^2 - 4$
- Q7.** If $f(x) = x - 3$ and $g(x) = x + 2$, what is $f(g(1))$?
(A) 0
(B) 2
(C) 1
(D) -1
(E) -2
- Q8.** What is the range of $f(g(x))$ if $f(x) = x^2$ and $g(x) = x - 1$?
(A) All real numbers
(B) All real numbers except $x = 0$
(C) $y \geq 0$
(D) $y \leq 0$
(E) $y = 0$

Open Ended Questions (FRQ)

- Q9.** Given $f(x) = \frac{1}{x}$ and $g(x) = x^2 - 1$, find $f(g(x))$ and determine its domain.
- Q10.** A water tank can hold x gallons of water. The amount of water in the tank after t hours is given by $f(t) = 100 - 5t$. The rate at which water is being pumped into the tank is $g(t) = 2t$. Find $f(g(2))$ and explain the result in the context of the problem.

Chef's Tips

- Ensure students have access to a calculator.
- Allow students to use a Composition of Functions table.
- Remind students to simplify their answers.